

LiveGuard AI

Multi-Modal Video Forensics & Integrity System

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1 Purpose of This Document

This document defines the complete functional, technical, and architectural requirements for LiveGuard AI, a production-grade video forensics system. The PRD is intended to serve as a single source of truth for reviewers, faculty, and technical evaluators to understand the system's scope, design decisions, machine learning methodology, and operational constraints.

2 Background and Motivation

Video has become the most trusted and persuasive form of digital evidence. With the rise of high-quality generative models and widespread content resharing, video is increasingly exploited to mislead audiences through:

- Facial manipulation and identity alteration
- Fully AI-generated synthetic videos
- Misrepresentation of real footage through false claims about time, place, or event

Human visual inspection is no longer sufficient to reliably assess such content, particularly under time pressure. Existing verification workflows are fragmented, reactive, and often opaque, creating a gap between content dissemination and credibility assessment.

3 Problem Statement

3.1 Core Problem

There is no unified, explainable system that assists users in assessing the credibility risk of video content across multiple manipulation dimensions at the time the content is being shared.

3.2 Key Challenges

- Modern manipulations are visually convincing and non-obvious
- AI-generated videos lack traditional camera artifacts
- Contextual misuse exploits trust in “real” footage
- Existing tools address only isolated aspects of the problem

4 Product Vision

LiveGuard AI is designed as an assistive forensic system that evaluates video content across three independent yet complementary analytical dimensions, producing risk-based, explainable outputs rather than definitive judgments.

5 Scope Definition

In Scope

- Short-form video analysis (10–20 seconds)
- Multi-modal forensic signals

- Explainable risk scoring
- GPU-accelerated inference
- Modular, extensible architecture

Out of Scope

- Truth determination
- Identity attribution
- Content enforcement
- Long-form live-stream ingestion (future extension)

6 High-Level System Capabilities

LiveGuard AI performs three primary analyses, presented in the following order:

1. Deepfake Risk Analysis
2. AI-Generated Video Detection
3. Context Integrity (Miscontext) Analysis

Each analysis can be executed independently on the same video input.

7 User Personas

7.1 Journalist / Media Verifier

- Needs rapid credibility signals
- Operates under time pressure
- Requires interpretability

7.2 Fact-Checking Analyst

- Performs deeper verification
- Needs evidence-backed signals

7.3 Platform or Research Analyst

- Reviews high volumes of content
- Uses risk scores for prioritization

8 User Workflow

1. User uploads a short video clip
2. User optionally provides an associated claim
3. User selects one or more analysis modes
4. System processes video and generates forensic signals
5. Results are presented with explanations and limitations

9 Functional Requirements

ID	Requirement
FR-1	Video Ingestion: Accept MP4, MOV formats; enforce duration and size constraints; validate codec compatibility
FR-2	Frame Extraction: Temporal sampling at fixed intervals; resolution normalization; artifact-preserving preprocessing
FR-3	Independent Analysis Execution: Each analysis module operates in isolation; failures in one module do not block others

10 Deepfake Risk Analysis Module

10.1 Objective

Assess whether a video contains facial manipulations or deepfake artifacts indicative of synthetic alteration.

10.2 Model Architecture Strategy

- Foundation vision encoder (CLIP-based)
- Parameter-efficient adaptation via normalization layers
- Metric-learning driven separation of real vs manipulated features

10.3 Training and Adaptation Narrative

The deepfake detector leverages pretrained models adapted from curated deepfake corpora derived from:

- FaceForensics++-style datasets
- Modern face swapping and reenactment methods
- Balanced real-fake pairs sharing source identity

Emphasis was placed on:

- Cross-manipulation generalization
- Reduction of dataset-specific shortcut learning
- Stability across unseen manipulation techniques

10.4 Inference Pipeline

1. Frame extraction
2. Feature embedding computation
3. Frame-level classification
4. Temporal aggregation
5. Confidence calibration

10.5 Output Signals

- Deepfake Manipulation Likelihood
- Temporal Consistency of Detection
- Model Confidence Stability

11 AI-Generated Video Detection Module

11.1 Objective

Detect whether a video is likely synthetically generated rather than camera-captured.

11.2 Theoretical Basis

AI-generated videos exhibit deviations in second-order temporal feature statistics that are difficult for humans to perceive but detectable algorithmically.

11.3 Model Strategy

- Video encoder for frame-level embeddings (XCLIP-based)
- Computation of covariance and higher-order statistics
- Training-free detection with statistical calibration

11.4 Calibration and Validation Narrative

The detector leverages pretrained encoders calibrated using a mixture of:

- Real-world camera footage
- AI-generated videos from diffusion and transformer-based generators
- Short-form video samples resembling social media content

11.5 Output Signals

- Synthetic Media Likelihood
- Temporal Feature Coherence
- Statistical Artifact Presence

12 Context Integrity (Miscontext) Analysis Module

12.1 Objective

Identify whether real video footage is being misrepresented via misleading claims.

12.2 Signals

- Temporal reuse detection
- Scene-claim semantic alignment
- Environmental consistency (lighting, time-of-day cues)

12.3 Methodology

- Frame embedding similarity search
- Scene-level classification
- Lightweight claim parsing

13 Risk Scoring Engine

13.1 Scoring Philosophy

The system produces risk indicators, not truth labels.

13.2 Aggregation

- Weighted signal combination
- Normalization to 0–100 scale

13.3 Risk Bands

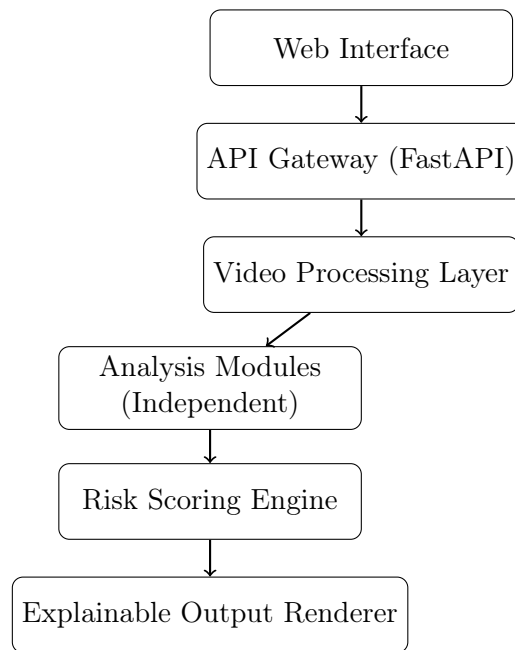
- LOW
- MEDIUM
- HIGH

14 Explainability Layer

Each result includes:

- Signal-level outcomes
- Human-readable explanations
- Executive summary
- Explicit uncertainty disclosure

15 System Architecture Overview



16 Performance Requirements

- End-to-end latency: < 20 seconds per clip
- GPU acceleration with CPU fallback
- Deterministic inference behavior

17 Security and Privacy

- Stateless processing
- No permanent storage
- No biometric identification
- Encrypted communication

18 Failure Modes and Mitigation

Failure Mode	Description	Mitigation
False positives	Over-flagging authentic content	Confidence bands + detailed explanations
Low-quality video	Degraded signal quality	Signal weighting and graceful degradation
Partial module failure	One analysis unavailable	Independent module execution

19 Ethical Considerations

- No automated enforcement

- Transparency-first design
- Human judgment remains final authority

20 Deployment Strategy

- Containerized backend
- Cloud-compatible GPU deployment
- Modular scaling per analysis module

21 Validation and Acceptance Criteria

- All three analyses operate independently
- Outputs are interpretable
- System remains stable under repeated use
- Results align with documented limitations

22 Future Extensions

- Audio-based manipulation analysis
- Live-stream sampling
- Cross-video coordination detection
- Browser-level integrations

23 Assumptions and Constraints

- Short-form videos only
- Limited reference corpus
- Near-live analysis focus

24 Final Summary

LiveGuard AI is a multi-modal, research-informed video forensics system that integrates deep-fake detection, synthetic media analysis, and contextual verification into a single, explainable platform. By prioritizing interpretability, modularity, and responsible deployment, the system demonstrates how advanced machine learning techniques can be applied to real-world credibility assessment without overclaiming or ethical compromise.